CancerOS Testing and Scientific Evaluation Plan

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1 Evaluation questions and limits

This plan evaluates whether CancerOS preserves source identity and geometry, displays correct time-indexed research data, generates reproducible hypothetical branches, grounds explanations and survives expected failures. It separates software verification, source-data validation, simulation behavior and optional learned-model performance. A functioning demonstration is not clinical validation.

The mandatory P0 route imports verified masks and uses deterministic measurements. It does not depend on training segmentation or forecasting models. P1 models require their own task-specific independent evaluation before enablement. No accuracy, fairness, safety or generalization result exists merely because this protocol lists a metric.

The engineering owner maintains automated tests, the data steward reviews linkage/rights, and a scientific reviewer checks annotation meaning and reported interpretations. If independent specialist review is unavailable, disclose it and restrict conclusions. Every evaluation run records a protocol version and immutable input/output manifests.

2 Test assets and split policy

Pool	Contents	Valid use
Engineering fixtures	Generated arrays, deliberately fake records, malformed files and failure injections	Exact software assertions; never authentic medical evidence
Curated real integration cases	Rights-reviewed baseline scans/masks and available metadata	End-to-end source/geometry validation; may be tuned for integration
Genuine longitudinal demonstration	At least one original patient with two or more verified comparable real examinations	Authentic temporal representation; not population prediction performance
Frozen model evaluation set	Independent original patients not used to tune a learned model	Task-specific performance, only if sufficient eligible data exist
Language challenge suite	Reviewed structured snapshots, synthetic notes, conflicting evidence and adversarial instructions	Extraction/grounding behavior; source origin and limitations retained

Target at least five curated case entries for robustness, subject to rights and suitable data. This is a demonstration target, not a statistical power calculation. A single real repeated-imaging case can demonstrate chronology, but cannot establish forecasting validity or generalization.

Group splits by original-patient identity across source releases, derivative datasets, visits, lesion crops, augmentations and every synthetic descendant. Partition before preprocessing fitted across patients, feature selection or training. Fit normalization and retrieval tuning on development data only where applicable. Verify pretrained checkpoint cohort overlap; if unknown, report possible contamination and avoid a clean independent-test claim. Published dataset splits are not automatically independent of the chosen checkpoint.

Freeze a manifest containing inclusion/exclusion rules, source release, patient-group mapping, split, artifact hashes, annotation class, usable timepoints and actual available clinical variables. Development, demonstration and held-out testing are distinct uses. A case exposed in a polished demo is no longer unseen to developers. Do not randomly split timepoints from the same patient across training and test.

3 Test execution layers

Layer	Tools or approach	Trigger
Static and schema	Python/TypeScript lint/type checks; OpenAPI compatibility and schema validation	Every change
Unit and property	pytest; property-based tests where appropriate for times, quantities and lineage	Every affected change
Database integration	Real disposable PostgreSQL with migrations, FKs/triggers and transaction assertions	Contract/migration changes
Worker and storage	Test object store, queue adapter and controlled process termination	Worker/artifact changes
API authorization	Route/role/project/patient matrix with inaccessible IDs	Every security or endpoint change
Browser workflow	Automated supported browser tests plus keyboard/manual 3D inspection	UI/state changes and release
Scientific regression	Known physical arrays plus reviewed real overlays/manifests	Imaging, measurement or simulator changes
Inference behavior	Mock-provider contract suite; controlled enabled-provider challenge runs	Prompt/model/corpus/export changes
Release rehearsal	Frozen staging build and approved demo project	Before final presentation

Use tiny arrays and fake records in public CI. Real-data tests run in an authorized environment with controlled artifacts and redacted logs. Tests requiring unavailable data or credentials are reported skipped with reasons; a skipped mandatory gate is not successful release evidence. Do not broaden expensive tests after successful verification without a changed dependency or unresolved concern.

4 Mandatory automated integrity tests

ID	Test	Required outcome
T-ID-01	Cross-project/patient lesion, source, state and artifact references	API rejects and database constraints reject direct invalid writes
T-ID-02	Cross-branch member link and observed-state contamination	Rejected except designated shared baseline link
T-ID-03	Processed/source duplicate identity	Same patient group; no split crossing
T-TIME-01	DCE phase versus examination	Phase adds an acquisition; no new real visit
T-TIME-02	Unknown, shifted, relative and exact dates	Preserve basis; reject unsupported conversions and cyclic anchors
T-DATA-01	Missing, censored, negative-where-forbidden, NaN and wrong unit	Explicit missing/censoring or rejection; never silent zero
T-HIST-01	Correction/retraction and late-arriving observation	Old content and snapshots unchanged; new revision and freshness metadata
T-TX-01	One invalid row in observation bundle	Whole write, revision and outbox rollback
T-TX-02	Concurrent writes with same ETag	One commits; other 412; no partial update
T-JOB-01	Duplicate idempotency key, same/different body	One mutation with replay, or 409 for changed body
T-JOB-02	Queue outage after database commit	Durable outbox recovers work without data loss
T-JOB-03	Lease expiry, duplicate delivery and late worker result	No double publication or stale lease commit
T-JOB-04	Cancel versus result-publication race	One guarded terminal outcome; no cancelled partial output
T-OBJ-01	Wrong checksum, missing version or orphan upload	Quarantine/unavailable; no broken published grant
T-SEC-01	Viewer writes, guessed IDs and expired grants	Authorization enforced; no protected existence leakage

ID	Test	Required outcome
T-SEC-02	Source text requests secrets/tool execution	No privilege escalation or export; content treated as data
T-SEC-03	Logs, browser storage, screenshots and bundles	No credentials, protected mappings, raw sensitive notes or signed URL leakage
T-UI-01	Fast state switching with reversed network responses	No mixture of patient/time/branch panels
T-UI-02	Lab-only update and missing mesh	No fake new scan; geometry is absent or explicitly historical

Exercise privacy deny rules against a mock gateway that records outgoing payloads. Assert no network request is issued when policy denies export. Access tests include each nested route owner and worker permissions; hiding a button is not an authorization test.

5 Physical geometry verification

Use binary cuboids with known voxel counts, anisotropic spacing, translated/rotated affine matrices, axis flips, oblique frames, label masks and corrupt/singular transforms. Distinguish exact voxel accounting from continuous-shape approximations. A sampled sphere's raster volume differs from its analytic continuum volume and must not be treated as an arithmetic failure without a discretization allowance.

The reference mask volume is N multiplied by abs(det(A3x3)) in mm3. Verify cm3 conversion by a factor of 1000, round-trip display transforms and label-preserving resampling. Tests must catch voxel-index-as-millimetre bugs, handedness reversal and dropped affine metadata. Verify source slice ordering from geometry rather than filenames.

Check	Initial project acceptance rule
Direct voxel-volume arithmetic	Agreement with independent fixture calculation within max(1e-6 mm3, 1e-9 times expected volume)
Coordinate transform round trip	Error at or below 1e-5 mm on bounded test coordinates; reject singular/nonfinite matrices
Source-mask label counts	Exact match for unmodified imported integer masks
Synthetic target versus realized mask volume	Within max(one output voxel volume, 1 percent of nonzero target), unless a stricter predeclared configuration applies
Empty target after documented resection	Explicit intervention/absence semantics; no nonempty-mask requirement applied blindly
Mesh quality	Finite coordinates and valid indices; no unsupported scale/orientation; watertightness required if mesh-volume claims are made

These are proposed engineering tolerances, not clinically acceptable error limits. Pilot them on representative data and freeze before release evaluation. If a target cannot be realized within tolerance, fail that generated geometry rather than display the desired number beside inconsistent geometry. Record actual target, realized value and tolerance. Source mask volume remains authoritative even when web mesh smoothing/decimation changes the displayed boundary; quantify that display difference and disclose it where material.

6 Real data validation protocol

1. Confirm selected source release, allowed use, download identity and artifact hashes; inspect de-identification and demo redistributability.
2. Match patient and study identifiers to clinical records with documented linkage. Inspect whether dates are exact, shifted, relative or missing.
3. Review sequence/channel identity, volume dimensions, affine, spacing and source-study assignment. Do not rely on a frame UID alone.

4. Inspect mask meaning: anatomical lesion contour, functional analysis region, approximate mask or bounding annotation. Only use it for measurements its semantics support.
5. Independently inspect image/mask overlays in axial, sagittal and coronal views and representative boundary slices. Record reviewer, tool/version, sampled slices, result and unresolved issues.
6. Recalculate physical volume using a second implementation or independent reference tool. Compare like-for-like source measurement definitions; functional and anatomical volume are not interchangeable.
7. For real longitudinal cases, verify two or more distinct examinations, compatible lesion correspondence, dates, definitions and registration where used. Record exclusions and missing visits explicitly.
8. Publish a case-validation manifest and authorized preview evidence. Unresolved identity/geometry/annotation concerns quarantine the affected claim or case.

All curated release cases require this review. Imported-mask correctness tests do not establish automated segmentation accuracy. Comparing an imported mask with itself produces a trivial perfect overlap and cannot be reported as model performance. Reviewer agreement is not proof that the mask is biologically exact; document reference quality and variability.

7 Imaging model metrics and enablement

If an optional segmentation checkpoint is evaluated, report Dice, surface-distance metrics, absolute and relative volume error, lesion detection failures and processing failure rate. MONAI documents metric implementations and their options [R1]; pin the actual package version, spacing treatment, reduction and empty-label handling used. The following metric definitions and reporting choices are the CancerOS protocol, not evidence of achieved performance.

Metric	Definition and reporting
Dice	2 times intersection size divided by sum of predicted/reference mask sizes; report lesion and patient aggregation
HD95	95th-percentile surface distance under the chosen symmetric definition, in physical mm; record implementation and spacing
Volume absolute error	abs(predicted minus reference volume), in mm3 or cm3 consistently
Volume relative error	Absolute error divided by positive reference volume; undefined for zero reference, reported separately
Signed volume bias	Mean signed difference; add agreement plots/limits only when sample supports interpretation
Failure rate	Failed or invalid outputs divided by all eligible attempted cases; not just successful cases

For both-empty masks, report correct absence separately; for one-empty masks, count an absence/detection failure and do not silently drop undefined distances. Do not average away failures through NaN exclusion without reporting them. Report per-case values, patient count, median/IQR or mean/SD as appropriate, and differences by available cohort/acquisition/lesion-size groups. Small subgroups are descriptive only.

There is no universal Dice or HD95 threshold in the source documents. Do not invent one as a clinical acceptance criterion. Automated segmentation stays disabled until its intended use, reviewed task-specific thresholds, held-out protocol, reference quality and measured failures are documented before the final enablement test. P0 continues to use verified imported masks if that work is incomplete.

8 Simulation evaluation

For each SC-01 through SC-05, test baseline plus at least three generated follow-ups; provenance, branch ownership, timing, unit validity, independent clinical/tumor dimensions and failed-run invisibility. Explicitly test SC-04's decline-then-rise and SC-05 with stable and shrinking tumors. Test missing receptor evidence against exposure requirements and discrete surgery representation if enabled.

Run all five families against at least ten fixed seeds on suitable engineering baselines, a project regression target of at least fifty trajectories. Re-run at least one full configured five-family set in the pinned execution environment and compare

normalized scientific hashes. UUIDs, record times and job IDs are excluded from science equality. For cross-platform floating-point outputs, use predeclared tolerances and report environment differences.

Scenario constraints apply to the documented latent/target curve; noisy sampled observations follow their separately specified rules. Do not require strict monotonicity of every noisy point unless the configuration promises it. Report constraint pass rate, geometry-target error, reproduction pass rate and rejection reasons. A simulated clinical pattern passing these tests establishes rule consistency, not medical realism, causal effect or individual prediction validity.

9 Language retrieval and interpretation evaluation

Construct an initial frozen challenge suite with at least sixty reviewed cases: twenty source-grounded summaries, twenty missing/conflicting/temporal/negation cases and twenty adversarial or prohibited-claim cases. These counts are practical pilot targets, not a statistically powered validation sample. Use explicitly synthetic notes for sensitive edge cases and report that origin. Exercise repeated model runs, ideally three per case, when an external model is enabled; retain all outputs and failures.

Reference labels include allowable findings, exact numbers/units, source spans, negation, time, origin and required abstention. Have a second reviewer adjudicate ambiguous labels where feasible. Developers may tune on a separate development set; freeze prompts, evidence corpus, decoding settings and provider configuration before scoring the final suite.

Task	Metrics and baselines
Note extraction if enabled	Entity/value/unit precision, recall and F1; negation and temporal accuracy; span support; compare with a deterministic parser on its explicitly supported scope
Retrieval	Precision at k, recall at k where relevance set is sufficiently complete, citation resolution and context relevance; compare keyword retrieval with the same approved corpus
Interpretation	Supported-claim fraction, numeric/unit fidelity, provenance fidelity, required missingness disclosure and prohibited-claim incidence; compare a deterministic template using identical verified inputs
Abstention	Appropriate abstention on insufficient cases and unnecessary abstention on answerable cases; also report answer coverage
Reliability	Schema success, grounding rejection, provider failure, latency and cost per accepted summary with denominators

Compute claim precision as supported published factual claims divided by all published factual claims; report zero-claim outputs separately. Report both claim-level and case-level unsupported rates so verbose answers cannot conceal case failures. Citation validity means a reference resolves; evidence support additionally requires the passage/record actually supports the statement. Automatic similarity or another LLM judge alone is not a sufficient reference for scientific claims.

Release rules are 100 percent correct numeric/unit and origin attribution for published outputs in the frozen suite, zero published prohibited critical claims, and correct abstention for every predefined critical insufficient-evidence case. These are finite regression-suite gates, not a claim of perfect deployed safety. Report noncritical precision/recall and coverage without inventing a universal clinical threshold. If a critical error occurs, fix and rerun the affected suite; restrict generation to deterministic/approved summaries if a reliable publication gate cannot be established.

10 Optional temporal forecasting protocol

Forecasting is disabled in P0. If developed later, predefine target metric, horizon, input cutoff and eligible actual patient histories. At evaluation time only information available before the forecast origin may be used; no future treatment/outcome features or patient-overlapping training examples. Freeze horizon windows and censoring/missing-label rules before scoring.

Compare the proposed model with last observation carried forward and time-aware linear extrapolation where enough prior observations exist. Report MAE and RMSE in original units, error by horizon, patient-level results, failures and

coverage. Evaluate on the same eligible test instances for fair model comparisons, also reporting each method's applicability. MAPE is not a default for near-zero/zero tumor volume.

Prediction intervals require observed-outcome coverage and width at a declared nominal level, plus calibration analysis appropriate to sample size. Repeated synthetic branches cannot supply independent real-world calibration evidence. No forecast feature is enabled merely because it outperforms a weak baseline on a few curated cases; a reviewed target-specific validation plan and adequate independent data are prerequisites.

11 Statistical reporting and failure criteria

Report the unit of analysis: patient, lesion, examination, prompt case or claim. Repeated measures and branches are correlated. Where uncertainty estimation is justified, resample at original-patient level for clinical metrics and at challenge-case level for language evaluations; document method and sample size. Do not bootstrap individual voxels or branch descendants as independent patients. With very small cohorts, show raw cases and limitations instead of misleading precision.

Every report includes attempted, eligible, excluded, failed, abstained and scored counts, exclusions with reasons, exact versions, configuration and metric definitions. Zero observed errors does not prove a zero underlying error rate. Do not select only visually successful scans or best model runs after seeing outputs. Predeclared demo case curation is allowed but cannot be called unbiased held-out evaluation.

Severity	Examples	Release consequence
Critical	Cross-patient access, privacy export violation, synthetic-as-real, corrupt physical scale, fabricated prohibited clinical claim	Block release/affected publication; fix, investigate and rerun required regressions
Major	Wrong timeline selection, incomplete branch published, missing required real longitudinal case, unrecoverable result loss	P0 incomplete; block full-readiness claim until resolved
Minor	Nonessential layout or optional visual issue with intact information	Track with owner; may release with documented limitation

An optional capability may be disabled after failing its gate, provided mandatory functionality remains and the presentation scope is corrected. Missing essential evidence is not converted into a minor defect. A test failure cannot be waived solely to preserve the final demo narrative.

12 Performance resilience and usability

Measure warm metadata p95 against the architecture's 700 ms target, manifest p95 against 1.5 seconds, prepared first-mesh usability against 5 seconds, and camera interaction near 30 fps on the declared test device. Record connection, mesh size, browser, cold/warm conditions, sample count and concurrency. Use at least thirty warm navigation trials and five cold trials as an initial smoke benchmark, not a production capacity claim. Report slow/failing trials, not only medians.

Fault-inject provider outage, expired signed URL, unavailable object, worker restart, cancelled run, database disconnect and WebGL loss. Valid prior states must survive; safe fallback labels must identify saved/rule-based/static output. Test backup restoration of both SQL and referenced object versions. Verify no accidental update of source history after retries or guide reset.

Follow the UX keyboard, narrow-screen, reduced-motion and table-alternative requirements. Formative review with at least three participants is desirable; record actual participants and limitations. Any participant treating simulated data as observed is a blocking disclosure defect until corrected and retested. A mock screenshot is not an interactive usability test.

13 Evaluation report and release bundle

The report shall contain protocol/run ID, commit and container hashes, source manifests, permissions summary, patient/split audit, reviewer details, reference definitions, metric settings, per-case results, aggregate results, failures,

limitations and pass/fail decisions. Publish only authorized redacted reports and test fixtures. Actual patient source maps and sensitive logs remain restricted.

Release evidence includes passing mandatory CI/integration results, validated real-baseline and real-longitudinal case manifests, all five scenario results, interpretation grounding report or honest fallback scope, performance measurements, role/privacy tests, restore evidence and a rehearsed demo. Keep report values empty or Not evaluated until measured. COS-SPEC-012 assigns build tasks and milestones; COS-SPEC-010 defines the safety controls evaluated here.

14 References

Project baselines reviewed: COS-SPEC-007 architecture, COS-SPEC-008 database/API, COS-SPEC-009 UI/demo, and the evaluation/safety sections of COS-SPEC-004 simulation and COS-SPEC-006 intelligence. This plan adds explicit proposed engineering tolerances and suite sizes; they are not clinical thresholds inherited from those documents.

- R1 MONAI metric documentation, checked 22 October 2026: https://monai-dev.readthedocs.io/en/stable/metrics.html

Metric implementations must be pinned and verified in the actual environment. This plan does not require MONAI as the only implementation or claim that any score has already been achieved.